

Faculty of Engineering - Ain Shams University
Electronics and Electrical Communications Engineering Department

ECE352s Analog and Digital Communication Systems

Junior - Electronics and Electrical Communications

Experiment (3)

Noise in Communication Systems and Sampling Theory

Spring 2026

1. Objectives:

- Observe the effect of Noise in communication systems.
- Quantify the performance of a communication system using signal-to-noise ratio.
- Sample and Reconstruct Analog Signals.

2. Equipment:

- Digital CRO
- Telecoms-Trainer 101 KIT

3. Part 1: Noise in AM Communication System

Any study of communications must inevitably include a consideration of noise. Noise is important because it interferes with the reproduction of the message at the receiver. For analog messages such as speech and music, noise is usually heard as hiss, crackle and pops that are superimposed on the recovered message.

The effects of noise must be objectively quantified. A widely used method for doing so is *signal-to-noise ratio* (SNR). As its name implies, SNR is simply the ratio of signal voltage (or power) to noise voltage (or power). The equation for calculating SNR is: $SNR = \frac{S}{N}$, Where: S= signal voltage or power; and N= noise voltage or power
It's better to measure the RMS value of signal and noise voltages for SNR calculations.

The greater the signal relative to the noise (which is desirable), the greater the SNR. That said, simple ratios can be misleading when dealing with signals and so decibels are often preferred.
The equation for calculating SNR as a decibel is either: $SNR(dB) = 20\log\frac{V_S}{V_N}$ or $SNR(dB) = 10\log\frac{P_S}{P_N}$

To perform these calculations, a measurement of the signal's voltage (or power) and the noise's voltage (or power) must be made. In most laboratory environments, the noise is controlled and can be removed from the signal. This makes separate measurements of signal and noise a relatively simple matter. However, in the field, the signal may not be available alone and so an alternative calculation must be used: $SNR = \frac{S+N}{N}$

Obviously, the two SNR ratio equations produce different numbers. However, for a simple quantitative comparison between two systems, this difference is irrelevant. Moreover, the greater the SNR of the system the smaller the difference between the values anyway.

For this experiment you'll use the Emona Telecoms-Trainer 101 to set-up an AM modulator. You'll then set-up an envelope detector that receives the AM signal via a bandwidth limited and noisy channel and make SNR measurements for three levels of channel noise.

Procedure

Part A - Setting up the AM modulator

1. Locate the Adder module and turn its G and *g* controls fully anti-clockwise.
2. Locate the Variable DCV module and turn its *DC Voltage* control almost fully anti-clockwise.
3. Connect the set-up shown in Figure 1.

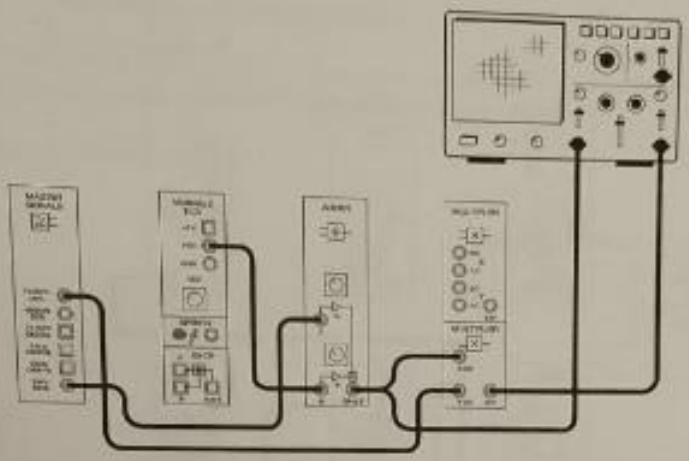


Figure 1

4. Use the scope's Channel 1 Vertical Position control to move the trace so that it lines up with the horizontal line in the middle of the scope's screen.
5. While watching the Adder module's output on the scope, turn its g control until the DC level is 1V.
6. While watching the Adder module's output on the scope, turn its G control to obtain a 1Vp-p sinewave.

The set-up in Figure 1 can be represented by the block diagram in Figure 2.
It generates a 100kHz carrier that is amplitude modulated (m=0.5 or 50%) by a 2kHz sinewave message.

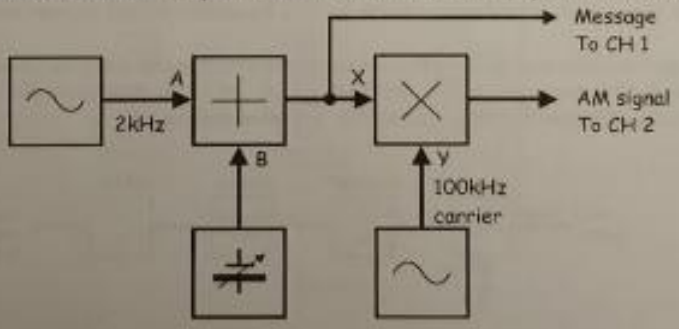


Figure 2

7. Adjust the scope's *Time base* control to view two or so cycles of the message.
8. Set the scope's Mode control to the DUAL position to view the AM signal out of the Multiplier module as well as the message signal.

Part B - Setting up the noisy channel and envelope detector

9. Modify the set-up as shown in Figure 3.

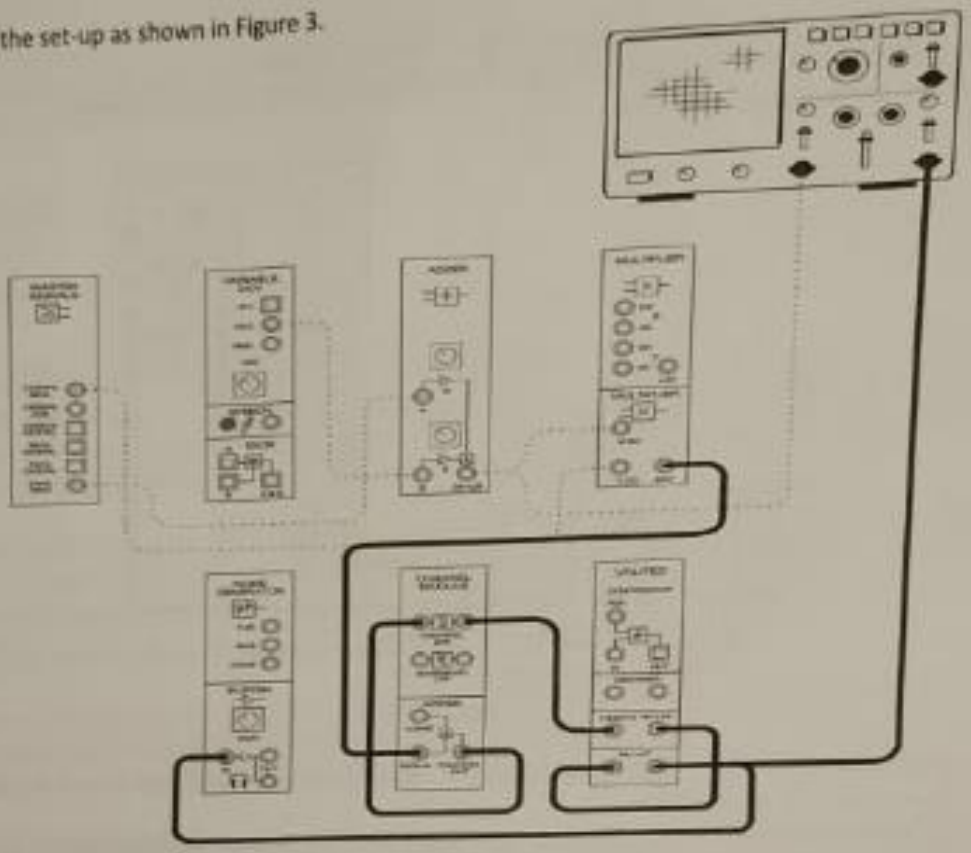


Figure 3

The set-up in Figure 3 can be represented by the block diagram in Figure 4.

The Adder and Channel BPF modules are used to model a bandwidth limited channel that can also be made noisy.
(Note: Although the Adder module's *NOISE* input is left unconnected, the input isn't *floating* due to the Telecoms-Trainer 101's hardware design). The Diode and RC LPF on the Utilities module implement an envelope detector while the additional RC LPF is used to filter any remaining carrier.

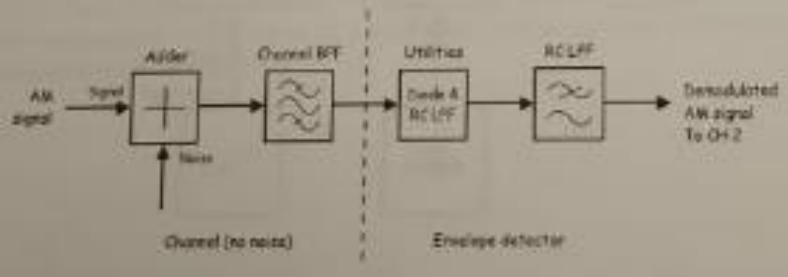


Figure 4

10. Set the scope's Channel 1 Input Coupling control to AC.
11. Compare the original message with the demodulated message.

Note: They should be the same shape and about the same size (though there will be a phase shift).

12. Turn the Buffer module's *Gain* control fully anti-clockwise to reduce it to minimum.
13. Without wearing the headphones, plug them into the Buffer module's headphone socket.
14. Put the headphones on.
15. Increase the Buffer module's gain until you can comfortably hear the recovered message.
16. Listen closely to the signal to see if you can hear any sound other than the 2kHz tone.

Part C - The effects of white noise on an envelope detector

Part C lets you see and hear the effects of white noise on the output of an envelope detector. You'll also quantify the noise problem for later comparison with product detection of AM signals.

17. Modify the set-up as shown in Figure 5.

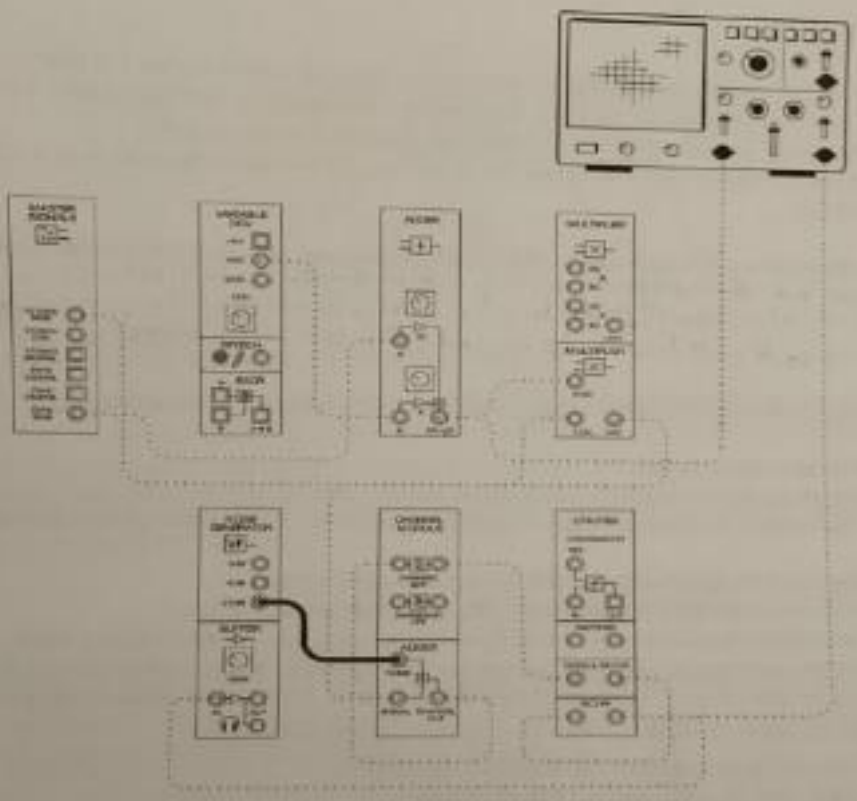


Figure 5

The set-up in Figure 5 can be represented by the block diagram in Figure 6.
The Noise Generator module models white noise at three levels: -20dB, -6dB and 0dB. As you can see, you'll i
start with -20dB which is the lowest level noise output.

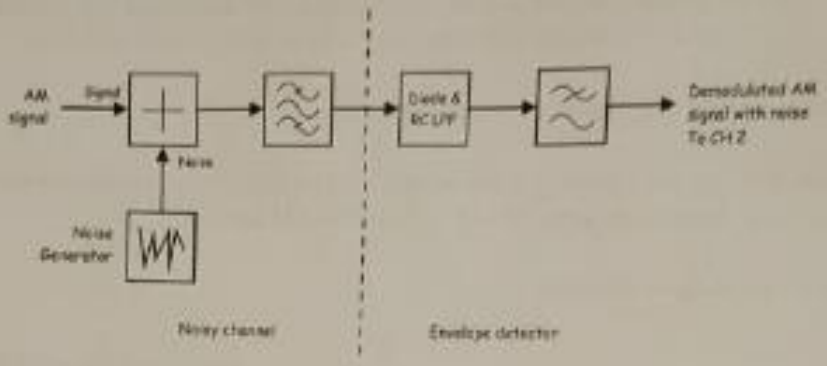


Figure 6

18. Look at and listen to the recovered message and note the effect on the signal's shape and sound.
19. Move the Adder module's NOISE input from the Noise Generator module's -20dB output to its -6dB output.
20. Note the effect of the increase in noise on the recovered message's shape and sound.
21. Switch the Adder module's NOISE input from the Noise Generator module's -6dB output to its 0dB output and repeat Step 20.

Why is the 2kHz tone still audible when -6dB and 0dB of introduced noise causes the signal's shape to be so distorted? WN distributes power accross wide bandwidth, while 2kHz tone concentrates its power at a signal spectral line ($f_0 = 2kHz$), the ear effectively increases SNR of 2kHz allowing the discrete tone to be resolved despite severe time domain.

The next part of the experiment gets you to quantify the noise problem at the receiver's output.

22. Return the noise level to -20dB.
23. Disconnect the patch lead between the Master Signals module's 2kHz SINE output and the Adder module's A input.
    Note: This removes the message from the input to the modulator and the output of the demodulator.
24. Set up the True RMS digital multimeter (DMM) for measuring AC voltages.
25. Connect the True RMS DMM between the RC LPF's output and the trainer's ground.
26. Measure the RMS voltage of the noise on the demodulator's output and record this value in Table.
27. Reconnect the patch lead between the Master Signals module's 2kHz SINE output and the Adder module's A input.
28. Determine the RMS voltage of the combined signal and noise on the demodulator's output and record this value.
29. Calculate and record the signal to noise ratio using the equation: $SNR = \frac{S+N}{N}$
30. Convert the SNR figure to decibels and record this value too.
31. Repeat Steps 23 to 30 for the -6dB and 0dB noise levels.

Noise Level	-20 dB	-6 dB	0 dB
Noise Voltage at output	0.0123	0.0442	0.0822
Signal + Noise Voltage	0.101V	0.111V	0.116V
SNR	9.92	2.5	1.34
SNR in dB	19	7.83	2.62

Part 2: Sampling and Reconstruction

Digital transmission is fast replacing analog in commercial communications applications. There are several reasons for this including the ability of digital signals and systems to resist interference caused by electrical noise, analog signal (like speech) must be converted to digital first. This involves sampling which requires that the analog signal's voltage be measured at regular intervals.

Figure 7a shows a pure sinewave for the message. Beneath the message is the digital sampling signal used to tell the sampling circuit when to measure the message. Beneath that is the result of "naturally" sampling the message at the rate set by the sampling signal. This type of sampling is "natural" because, during the time that the analog signal is measured, any change in its voltage is measured too.

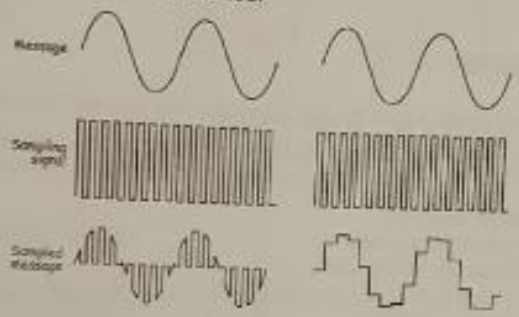


Figure 7a Figure 7b

For some digital systems, a changing sample is unacceptable.

Figure 7b shows an alternative system where the sample's size is fixed at the instant that the signal measured. This is known as a sample-and-hold scheme (and is also referred to as pulse amplitude modulation). Regardless of the sampling method used, by definition it captures only pieces of the message. So, how can the sampled signal be used to recover the whole message? This question can be answered by considering the mathematical model that defines the sampled signal:

$$\text{Sampled message} = \text{the sampling signal} \times \text{the message}$$

As you can see, sampling is actually the multiplication of the message with the sampling signal. And, as the sampling signal is a digital signal which is actually made up of a DC voltage and many sinewaves (the fundamental and its harmonics) the equation can be rewritten as:

$$\text{Sampled message} = (\text{DC} + \text{fundamental} + \text{harmonics}) \times \text{message}$$

When the message is a simple sinewave (like in Figure 7) the equation's solution (which necessarily involves some trigonometry that is not shown here) tells us that the sampled signal consists of:

- A sinewave at the same frequency as the message
- A pair of sinewaves that are the sum and difference of the fundamental and message frequencies
- Many other pairs of sinewaves that are the sum and difference of the sampling signals' harmonics and the message

This ends up being a lot of sinewaves but one of them has the same frequency as the message. So, to recover the message, all that need be done is to pass the sampled signal through a low- pass filter. As its name implies, this type of filter lets lower frequency signals through but rejects higher frequency signals.

That said, for this to work correctly, there's a small catch which is discussed in Part B of the experiment.

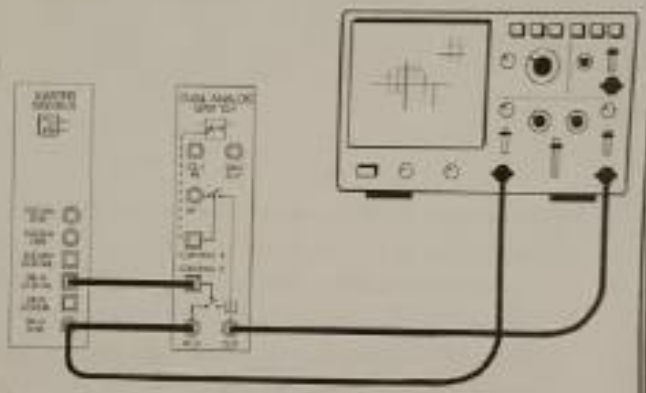
In this experiment you'll use the Emona Telecoms-Trainer 101 to sample a message using natural sampling and a sample-and-hold scheme. You'll then reconstruct the message from the sampled signal and examine the effect of aliasing.

Part A - Sampling a simple message

The Emona Telecoms-Trainer 101 has a Dual Analog Switch module that has been designed for sampling. This part of the experiment lets you use the module to sample a simple message using two techniques.

1. Connect the set-up shown in Figure 8.
2. Adjust the scope's *Time base* control to view two or so cycles of the Master Signals module's *2kHz SINE* output.
3. Set the scope's *Mode* control to the *DUAL* position to view the sampled message out of the Dual Analog Switch module as well as the message.
4. Draw the two waveforms to scale in the space provided.

   Leave some space for a 3rd signal.



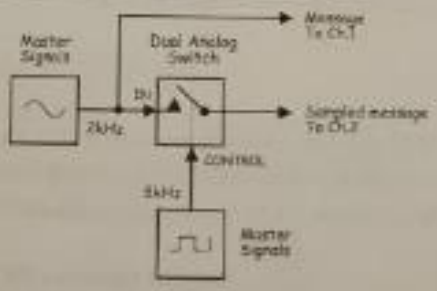
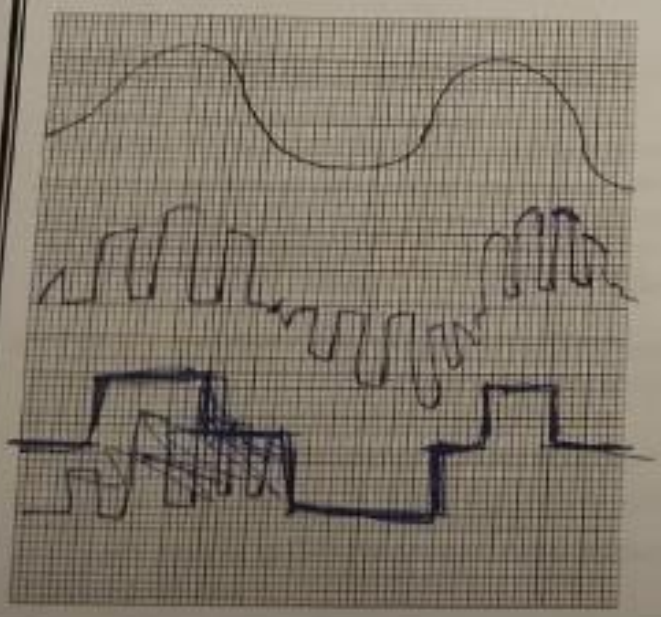


Figure 8



5. Modify the set-up as shown in Figure 9.

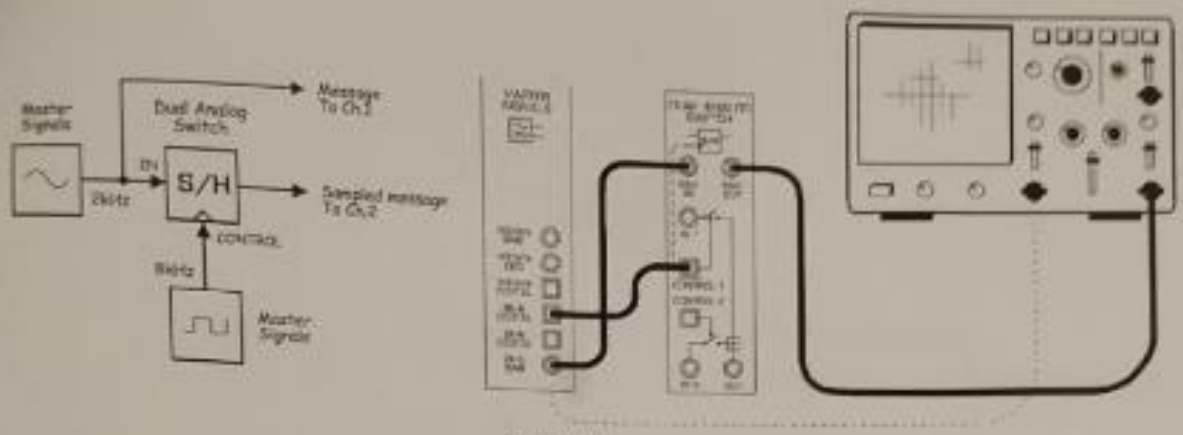


Figure 9

6. Draw the new sampled message to scale in the space that you left on the graph paper.

Part B - Reconstructing a sampled message

Recall that the sampled message is made up of many sinewaves. Importantly, for every sinewave in the message, there's a sinewave in the sampled message with the same frequency. So "reconstructing" the original message involves passing the sampled message signal through a low-pass filter. This lets the sinewave (or sinewaves) with the same frequency as the message through while rejecting the other sinewaves.

7. Locate the Tuneable Low-pass Filter module and set its Gain control to about the middle of its travel.
8. Turn the Tuneable Low-pass Filter module's Cut-off Frequency Adjust control fully anti- clockwise.
9. Modify the set-up as shown in Figure 10.

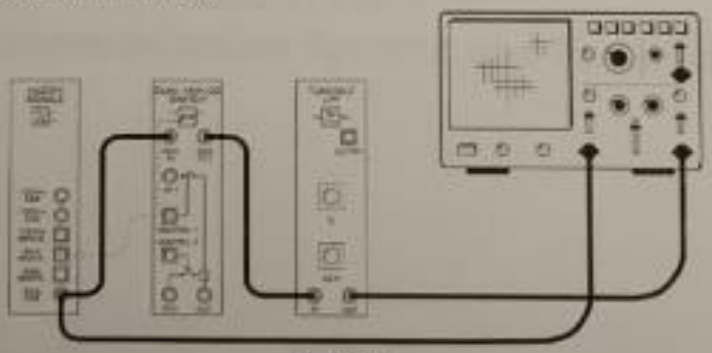


Figure 10

At this point there should be nothing out of the Tuneable Low-pass Filter module. This is because it has been set to reject almost all frequencies, even the message. However, the cut- off frequency can be increased by turning the module's Cut-off Frequency Adjust control clockwise.

10. Slowly turn the Tuneable Low-pass Filter module's Cut-off Frequency control clockwise and stop the moment the message signal has been reconstructed.

Plots of the waveforms from the lab:

